

Causal Inference in Epidemiology

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Last modified: 2026-01-27: 5:34:35 (UTC)

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Preface

This will be an online book about causal inference.

Here are some other resources for learning causal inference:

UC Davis courses

- EPI 205¹ “Principles of Epidemiology”
- EPI 206² “Epidemiologic Study Design”
- EPI/SPH 207³ “Advanced Epidemiologic Methodology”
- EPI 225⁴ “Advanced Topics in Epidemiology Methods”
- POL 285⁵ “Statistics of Causal Inference in Political Science”
- MGB/MGP/MGT 454A⁶ “Causal Inference and Statistical Experiments”
 - syllabus: <https://webapps.aws.ucdavis.edu/public/documents/4861649/Syllabus><https://schedule.aws.ucdavis.edu/public/documents/5319910/Syllabus>
- PSC 204B⁷ “Causal Modeling of Correlational Data”
- PSC 205C⁸ “Structural Equation Modeling”

Course search options:

- <https://schedule.aws.ucdavis.edu/courseScheduling>
- <https://catalog.ucdavis.edu/course-search/>
- <https://catalog.ucdavis.edu/courses-subject-code/>

Online Videos

- “Introduction to Causal Inference”⁹ (slides here¹⁰)
- Online Causal Inference Seminar series¹¹

¹<https://catalog.ucdavis.edu/search/?q=EPI+205>

²<https://catalog.ucdavis.edu/search/?q=EPI+206>

³<https://catalog.ucdavis.edu/search/?q=EPI+207>

⁴<https://catalog.ucdavis.edu/search/?q=EPI+225>

⁵<https://catalog.ucdavis.edu/search/?q=POL+285>

⁶<https://catalog.ucdavis.edu/search/?q=MGB+454A>

⁷<https://catalog.ucdavis.edu/search/?q=PSC+204B>

⁸<https://catalog.ucdavis.edu/search/?q=PSC+205C>

⁹<https://ucdhs.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=43e9eb6f-3ed9-41ac-8ad9-ae22016572c8%20>

¹⁰<https://health.ucdavis.edu/media-resources/ctsc/documents/pdfs/causal-inference-intro-2022.pdf>

¹¹<https://www.youtube.com/channel/UCiiOj5GSES6uw21kfXnxj3A/videos>

UC Davis Datalab learning group

- <https://datalab.ucdavis.edu/causal-inference/>
 - Reading list¹²

Other links:

- <https://cameron.econ.ucdavis.edu/causal/>
- https://datalab-icmat.github.io/causal_reading_group.html
- Lab exercises by Ben Noble¹³: <https://github.com/bennoble/causal-inference-2022>

Books

- Judea Pearl (2016)
- Hernán and Robins (2020)

¹²https://docs.google.com/document/d/1K0QZFSjQIYnOTahpRK7Q83eaiIIFjfa-clSbj_ifgco/edit?tab=t.0#heading=h.farbmh6n76gq

¹³<https://benjaminncoble.org/>

1 Introduction to causal inference

1.1 Introduction

Hernán and Robins (2020) (page vii):

Unfortunately, the scientific literature is plagued by studies in which the causal question is not explicitly stated and the investigators' unverifiable assumptions are not declared. This casual attitude towards causal inference has led to a great deal of confusion.

Rigorously defining cause and effect is difficult. Fortunately, many humans have strong intuitions about these concepts. We will make cursory attempts at definitions for the basic terms, and leave the finer points to philosophers.

1.2 Individual causal effects

Definition 1.1 (Action, intervention, exposure, policy, treatment). An **action** (also called an **intervention**, **exposure**, **policy**, or **treatment**) is a choice that we consider making.

Definition 1.2 (Random variable). A **random variable** is a variable that may have different values for different individuals and/or for different actions/exposures that an individual might experience. ¹

Definition 1.3 (potential outcome, consequence). A **potential outcome** of action a (also called a **consequence** of a) is the value of a random variable Y that would occur if we were to take action a . The potential outcome of action a on random variable Y is often denoted $Y(a)$, Y^a , or $Y^{A=a}$. We will use notation $Y(a)$.

¹adapted from Hernán and Robins (2020) p3

Definition 1.4 (Factual outcome, observed outcome). A *factual outcome* (or *observed outcome*) is the potential outcome corresponding to an action that was actually taken.

If we consider taking action a or an alternative a' , and we actually take action $A = a$, then $Y(a)$ is the factual outcome.

Definition 1.5 (Counterfactual outcome). A counterfactual outcome is a potential outcome corresponding to an action that was not actually taken.

If we consider taking action a or an alternative a' , and we actually take action $A = a$, then $Y(a')$ is a counterfactual outcome.

There might be more than one counterfactual outcome, depending on how many action options were considered, but there can only ever be one factual outcome per random variable.

Definition 1.6 (Cause). Action a **causes** outcome y (or is a cause of y) if:

- outcome y would occur if we were to take action a
and
- outcome y would not occur if we did not take action a .

In other words, if:

- $Y(a) = y$
and
- $\exists a' \neq a : Y(a') \neq y$

Definition 1.7 (effect). The effect of action a on outcome Y , relative to a given alternative action a' , is the contrast in potential outcomes, $Y(a)$ versus $Y(a')$.

Definition 1.8 (Consistency). Consistency is the assumption that if we observe an action a , then the observed outcome Y is equal to the “factual potential outcome” corresponding to action a ; in other words, if $A = a$, then $Y(a) = Y$.

Definition 1.9 (Exchangeability). Subpopulations defined by exposure X are exchangeable with respect to a potential outcome $Y(x)$ if the distribution of $Y(x)$ does not depend on the observed exposure X :

$$Y(x) \perp\!\!\!\perp X$$

Theorem 1.1. *If subpopulations defined by values of exposure X are exchangeable with respect to potential outcome $Y(x)$, then the expected value of $Y(x)$ does not depend on the observed value of X :*

$$E[Y(x)|X = x'] = E[Y(x)|X = x]$$

Definition 1.10 (Conditional exchangeability). Subpopulations defined by exposure X are exchangeable with respect to a potential outcome $Y(x)$ if the distribution of $Y(x)$ does not depend on the observed exposure X , conditional on covariate(s) Z :

$$Y(x) \perp\!\!\!\perp X | \tilde{Z}$$

Theorem 1.2. *If subpopulations defined by values of exposure X are conditionally exchangeable with respect to potential outcome $Y(x)$ given covariate $\tilde{Z}$, then the expected value of $Y(x)$ does not depend on the observed value of X :*

$$E[Y(x)|X = x', \tilde{Z} = \tilde{z}] = E[Y(x)|X = x, \tilde{Z} = \tilde{z}]$$

Definition 1.11 (Fundamental problem of causal inference). The *fundamental problem of causal inference* is that only one potential outcome (the factual outcome) can be observed per person (or per sampling unit, more generally) (Holland 1986). The other, counterfactual outcomes, are all missing data, and thus, the causal effects are all missing data as well.

1.3 Common biases in causal inference

1.3.1 Collider bias

Collider bias is a type of selection bias that occurs when conditioning on a common effect (collider) of both an exposure and an outcome. This conditioning can create a spurious association between the exposure and outcome even when no true causal relationship exists in the general population.

A collider is a variable that is caused by both the exposure and the outcome of interest. When we condition on this variable (through study design, selection criteria, or statistical adjustment), we can introduce bias. For example, if both Disease A and Disease B independently increase the likelihood of hospitalization, then analyzing only hospitalized patients can create a false association between these diseases.

The Catalog of Bias provides comprehensive information about collider bias and many other biases that can affect research validity. The catalog includes:

- Clear definitions and explanations
- Real-world examples from health research
- Directed acyclic graphs (DAGs) illustrating bias mechanisms
- Guidance on study design and analysis to avoid or minimize bias

For detailed information about collider bias, see: <https://catalogofbias.org/biases/collider-bias/>

1.4 For a complete catalog of biases in health research, visit: <https://catalogofbias.org/>

“Continuation of Annual Screening Mammography and Breast Cancer Mortality in Women Older Than 70 Years” (2020), supplementary materials p16:

A sensitivity analysis for unmeasured confounding can only conclude that, if there is much unmeasured confounding, there will be much bias in the estimates and, conversely, if there is little unmeasured confounding, there will be little bias in the estimates. But, if the magnitude of unmeasured confounding is unknown (as it often is), then the magnitude of confounding bias is also unknown. That is, sensitivity analysis for unmeasured confounding is, by definition, uninformative. The situation is different when investigators have some information about the unmeasured confounder(s). For example, our analysis did not adjust for cigarette smoking, which may be associated with both breast cancer screening (because smokers are less likely to use preventive services) and breast cancer death. If we knew the prevalence of smoking in screening-defined groups and the association between smoking and breast cancer death, then we could use published methods to correct the confounded effect estimates.

Unfortunately, these methods assume that the residual confounding can be summarized by a single variable that happens to be dichotomous, time-fixed, and unassociated with all measured variables that were adjusted for in the analysis (Lash, Fox, and Fink 2009; Schneeweiss 2006).

See also Fox, MacLehose, and Lash (2022) for an updated reference.

2 Difference in differences analyses

Many approaches to causal inference assume exchangeability (Definition 1.10) and exploit its consequence (Theorem 1.1):

$$\mathbb{E}[Y(x)|X = x'] = \mathbb{E}[Y(x)|X = x]$$

Difference-in-differences makes a weaker exchangeability assumption:

$$\mathbb{E}[Y_t(0) - Y_{t'}(0)|X = 1] = \mathbb{E}[Y_t(0) - Y_{t'}(0)|X = 0]$$

3 Summary

In summary, this book has no content whatsoever.

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References

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